

ID4802

OELP Project Report

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Comprehensive setup for temperature, humidity, and gas sensor fusion for Indoor Air Quality (IAQ)

1. Problem Statement & Objectives

People spend nearly 90% of their time indoors, yet indoor air quality remains largely invisible and unmonitored. Pollutants such as CO₂, volatile organic compounds (TVOCs), and fine particulate matter (PM_{2.5}, PM₁₀) accumulate silently and are directly linked to respiratory disease, fatigue, and reduced cognitive function. Existing monitoring systems are either expensive and require professional installation, or measure only a single parameter with no standardized AQI output.

This project implements a low-cost, IoT-based IAQ monitoring system that measures air quality parameters, computes India's CPCB-standard AQI in real time, and makes the data accessible on a live cloud dashboard from any device anywhere.

Objectives

1. Design a real-time multi-parameter IAQ monitoring system measuring CO₂, TVOC, PM_{1.0}, PM_{2.5}, PM₁₀, Temperature, and Humidity.
2. Implement India's CPCB AQI calculation using standard breakpoint tables for PM_{2.5}, PM₁₀, CO₂, and TVOC on a 0–500 scale.
3. Transmit sensor data continuously to the cloud over 4G LTE cellular, independent of Wi-Fi infrastructure.
4. Store every reading in a PostgreSQL time-series database enabling historical analytics and date-range filtering.
5. Display live readings and trend charts on an interactive React dashboard accessible from any browser.

2. AQI Calculation Methodology

The Air Quality Index (AQI) is a dimensionless scale from 0 to 500 defined by India's Central Pollution Control Board (CPCB). A sub-index is computed for each pollutant using linear interpolation between concentration breakpoints. The final AQI is the maximum sub-index, and the pollutant driving that maximum is called the dominant pollutant.

$$I_p = [(I_{\text{High}} - I_{\text{Low}}) / (BP_{\text{High}} - BP_{\text{Low}})] \times (C_p - BP_{\text{Low}}) + I_{\text{Low}}$$

$$\text{Final AQI} = \max(I_{\text{PM2.5}}, I_{\text{PM10}}, I_{\text{CO}_2}, I_{\text{TVOC}})$$

AQI Range	Category	PM2.5 (µg/m³)	PM10 (µg/m³)	CO ₂ (ppm)	TVOC (µg/m³)
0 – 50	Good	0 – 30	0 – 50	0 – 600	0 – 65
51 – 100	Satisfactory	31 – 60	51 – 100	601 – 1000	66 – 220
101 – 200	Mod. Polluted	61 – 90	101 – 250	1001 – 1500	221 – 660
201 – 300	Poor	91 – 120	251 – 350	1501 – 2000	661 – 2200
301 – 400	Very Poor	121 – 250	351 – 430	2001 – 5000	2201 – 5500
401 – 500	Severe	> 250	> 430	> 5000	> 5500

3. Hardware Components & Connections

The system is built around the ESP32-S3 microcontroller, which manages all sensor communication, AQI computation, and 4G LTE uplink. Three sensor modules are connected across two I²C buses and one UART port.

Module	Interface	Pins	Parameters
SCD4x	I ² C Bus 1	SDA=2, SCL=1	CO ₂ , Temp, Humidity
ENS160	I ² C Bus 1	SDA=2, SCL=1	TVOC, eCO ₂
PMS5003	UART2	RX=42, TX=41	PM1.0, PM2.5, PM10
EC200U	Serial1	TX=13, RX=12	4G LTE uplink

Firmware Logic

The ESP32 firmware runs a 5-second main loop. Each iteration reads all sensors, pushes values into 6-sample circular rolling average buffers, computes sub-indices, and determines the dominant pollutant. Every 20 seconds it serialises the averages into a JSON payload and publishes to EMQX via AT+QMTPUB. Rolling averages smooth out momentary sensor noise before the data reaches the cloud.

4. MQTT & Cloud Pipeline

Sensors → ESP32-S3 → EC200U (4G LTE) → EMQX Broker → Bridge Script → Supabase → Dashboard

Stage	Technology	Detail
1 . Publish	EC200U + AT Commands	ESP32 activates PDP context on Airtel GPRS, opens MQTT session via AT+QMTOPEN, publishes JSON every 20s via AT+QMTPUB on topic iaq/palakkad/datamos
2 . Broker	EMQX (broker.emqx.io:1883)	A public MQTT broker receives the payload and routes it to all subscribers. Protocol: MQTT v4, QoS 0, Client ID: ESP32-IAQ-01
3 . Bridge	Node.js on Render.io	Subscribes to the topic, parses JSON, inserts a row into Supabase iaq_readings using the service_role key. Runs 24/7 and data is captured even when no browser tab is open
4 .Database	Supabase PostgreSQL	Stores every reading permanently. Supabase Realtime pushes each new INSERT to connected WebSocket clients instantly with no polling required
5 . Dashboard	React + Vercel	On load fetches historical rows for charts. Realtime WebSocket updates live tiles and extends chart by one point every 20s

JSON Payload Structure

Each publish contains: ts (timestamp), aqi, dominant, CO₂, tvoc, pm1, pm25, pm10, temp, hum, si_co2, si_tvoc, si_pm25, si_pm10. All values are rolling averages computed on-device. Sub-indices for CO, NO₂, NH₃ are also included for reference but excluded from the final AQI.

5. Database - Supabase (PostgreSQL)

Without a database the dashboard can only display the current reading . The moment the next one arrives, the previous is lost forever. Supabase provides persistent storage for every 20-second reading, enabling the core analytics use case of this project.

Why Supabase

- Every reading is permanently stored and answers questions like “what was AQI at 2pm yesterday?” or “which hour is consistently the worst?”
- Realtime engine broadcasts each new INSERT to the dashboard via WebSocket in under one second
- Auto-generated REST API (PostgREST) frontend queries historical data with no custom backend code
- Row Level Security with public read, authenticated write; secret service_role key stays in the bridge, never in the browser

Table Schema - iaq_readings

Column	Type	Description
id	BIGSERIAL PK	Auto-incrementing unique row ID
ts	BIGINT	Unix timestamp from ESP32
aqi	INTEGER	Final AQI (0–500)
dominant	TEXT	Dominant pollutant name
co2	REAL	CO ₂ ppm
tvoc	REAL	TVOC ppb
pm1 / pm25 / pm10	REAL	Particulate matter µg/m ³
temp / hum	REAL	Temperature °C and Humidity %RH
si_co2 / si_tvoc / si_pm25 / si_pm10	INTEGER	AQI sub-indices per pollutant
created_at	TIMESTAMPTZ	Server-side insert timestamp

6. Frontend Dashboard

Tech Stack

- React + Vite - component-based UI, fast build system
- Supabase JS Client - historical fetch + Realtime WebSocket subscription
- Recharts - SVG line charts rendered entirely in the browser
- Tailwind CSS + Glassmorphism - dark frosted-glass card styling
- Deployed on Vercel at <https://iaqiot.vercel.app>

How Data Flows to the Dashboard

Two mechanisms run simultaneously when the dashboard loads:

1. On page load , the app queries Supabase for the last N rows ordered by timestamp. This data populates the historical Recharts line chart and all sensor tiles with the latest reading.
2. Realtime subscription : A WebSocket channel listens for INSERT events on `iaq_readings`. Every time the bridge inserts a new row, Supabase pushes it to the frontend instantly. React updates component state and tiles refresh and Recharts appends one new point to the right of the chart. The graph literally grows live every 20 seconds with no page reload.

Dashboard Features

- Live AQI number with colour-coded category badge (Good → Severe)
- Dominant pollutant indicator and shows which parameter is driving the AQI
- 6 sensor tiles which include CO₂, TVOC, PM2.5, PM10, Temperature, Humidity
- Historical AQI line chart via Recharts with date range filter
- Zero polling which means all live updates driven by Supabase Realtime WebSocket
- Fully responsive so that works on desktop and mobile
- An on-dashboard alert system that displays a visible warning banner when the AQI rises above the Satisfactory threshold..

7. Conclusion

This project successfully implements a complete end-to-end indoor air quality monitoring system using low-cost commercially available sensors, an ESP32-S3 microcontroller, 4G LTE cellular connectivity, and a modern cloud-native software stack.

The system continuously measures seven air quality parameters, computes India's CPCB AQI every 20 seconds, and transmits it to the cloud over cellular requiring no Wi-Fi infrastructure. The bridge script running 24/7 on Render.io guarantees no data is ever lost. Supabase provides both permanent storage for historical analytics and real-time push for the dashboard. The React frontend delivers live readings and trend charts accessible globally from any device.

4 Sensor Modules	11 Parameters	20s Upload Cycle	AQI 0–500
SCD4x, ENS160, PMS5003, EC200U	CO ₂ , TVOC, PM1/2.5/10, Temp, Hum + sub-indices	Continuous 24/7 cellular cloud logging	India CPCB standard, 6 categories

Future Scope

- Automated SMS/email alerts when AQI crosses a user-defined threshold
- Machine learning models for AQI prediction from historical patterns
- Multi-location deployment with a unified monitoring dashboard and sensor fusion
- Mobile app with push notifications and personalized health advisories